



LAND OF THE CURIOUS





LECTURE 1 – PRODUCT DEVELOPMENT FUNDAMENTALS

CS39A0040 Product & Service Development

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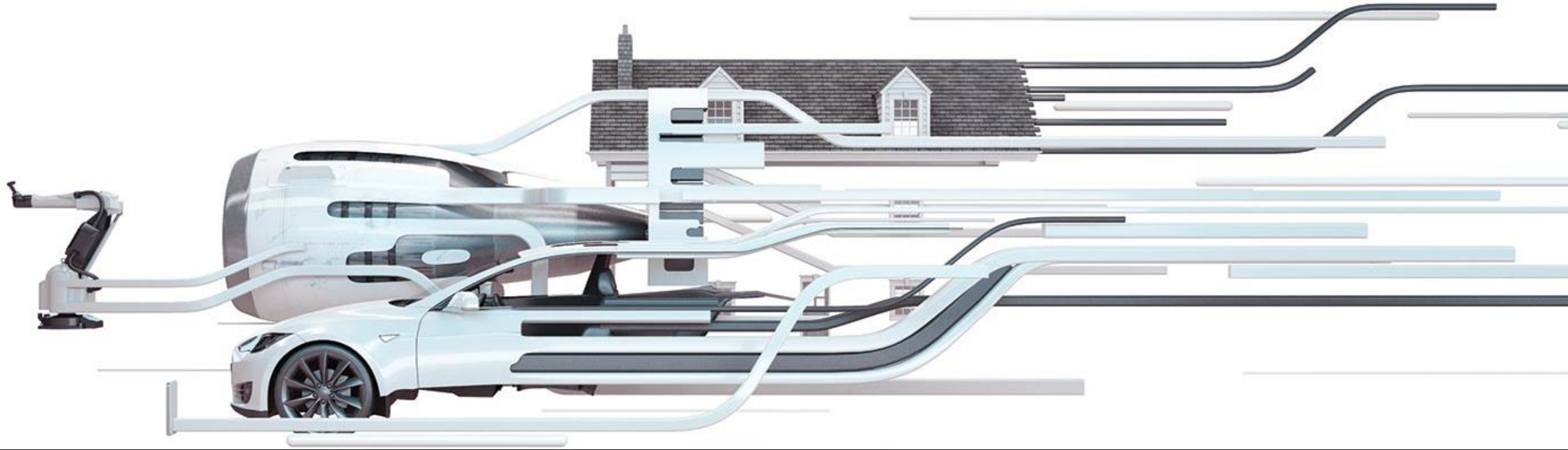


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*Business is like a car: it will not run by itself
except downhill.*

American Saying





INTRODUCTION AND BACKGROUND

Product, Service, Product Service System



CONTENT

1. Introduction and Background
2. Product-Service Systems (PSS)
3. Product Lifecycle Management (PLM)
4. Digital Platforms
5. Digital Twins
6. Enabler Technologies (Realtime Simulation, IoT, IMS, connectivity, intelligence/smart)

PRODUCT & SERVICE DEVELOPMENT

Product Lifecycle Manage. (PLM)

- Product Lifecycle Management (PLM) is:
- the business activity of managing a company's products and services all the way across their lifecycles
 - an integrative information-driven management approach

Product

- A Product is:
- A value proposition a business network offers to or co-produces with clients
- It consists of a mix of tangible **products** and intangible **services** designed and combined so they jointly fulfil customer needs

Service

- A Service consists of:
- Leveraging a company's specialised knowledge and skills through activities, behaviours & processes
 - Results in an outcome in customers assets, processes and operations.

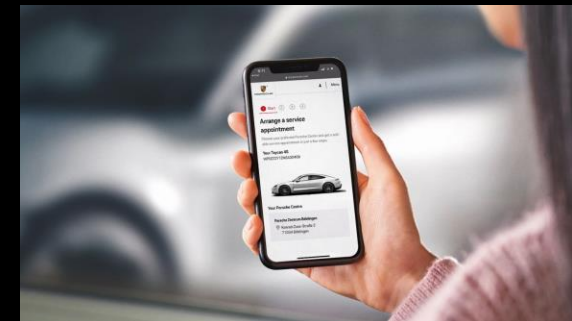
Product-Service System

- PSS is:
- a market proposition to customers, users, stakeholders
 - extends the traditional functionality of a product by incorporating additional services

Digital Twin

- A physics-based or data description of the real-world system resulting from:
1. generation,
 2. management,
 3. application of information from sources across the system's lifecycle.

PRODUCTS, SERVICES AND PRODUCT-SERVICE SYSTEMS



COMMON VIEW OF PRODUCTS AND SERVICES

1. Simple product (Equipment)?
2. Complex solution (MP Plant)
3. Asset based services
4. Product-Service-System (integrated product and services)
5. intelligent Product-Service-Systems (lifecycle ecosystems)



INTELLIGENT PRODUCT-SERVICE SYSTEM IDEA

Smart Connect Product System
(Real-world Domain)



Decision Making in operational
lifecycle

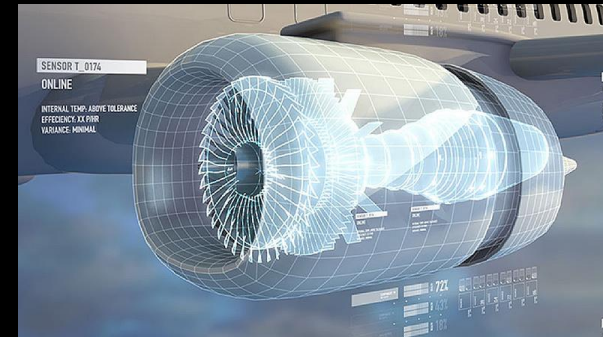
Information



Data

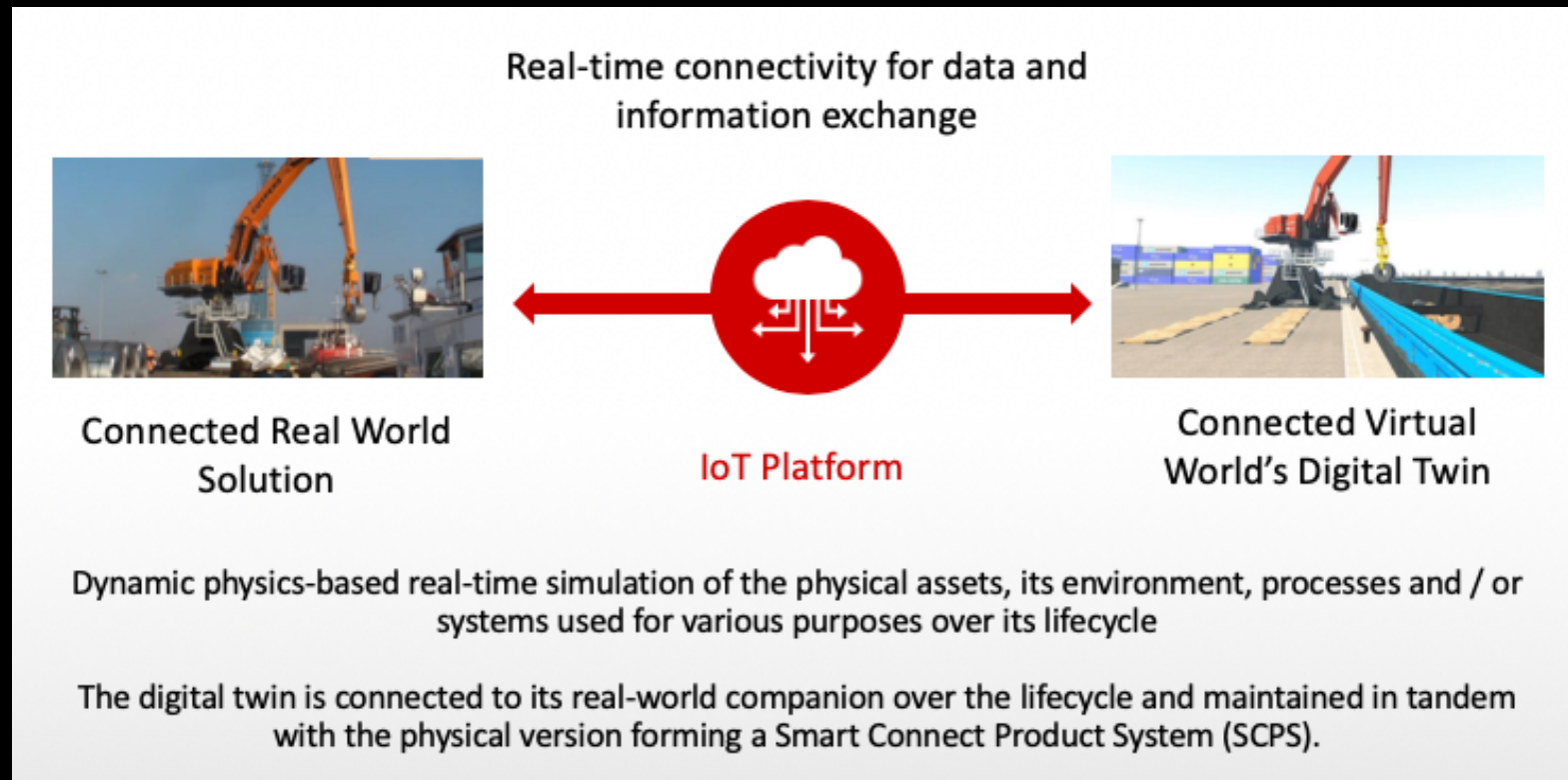


Smart Connected Digital
Twin (Virtual Domain)

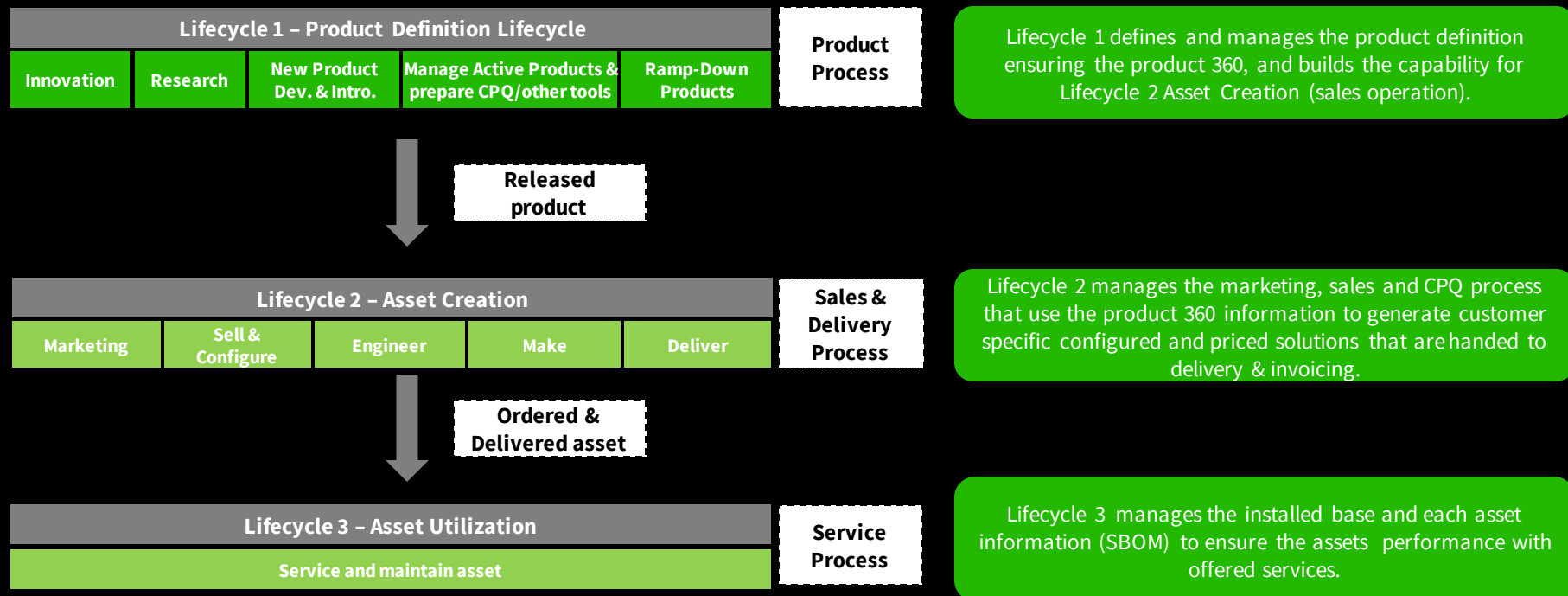


Decision Making over the
lifecycle

SMART CONNECTED PSS

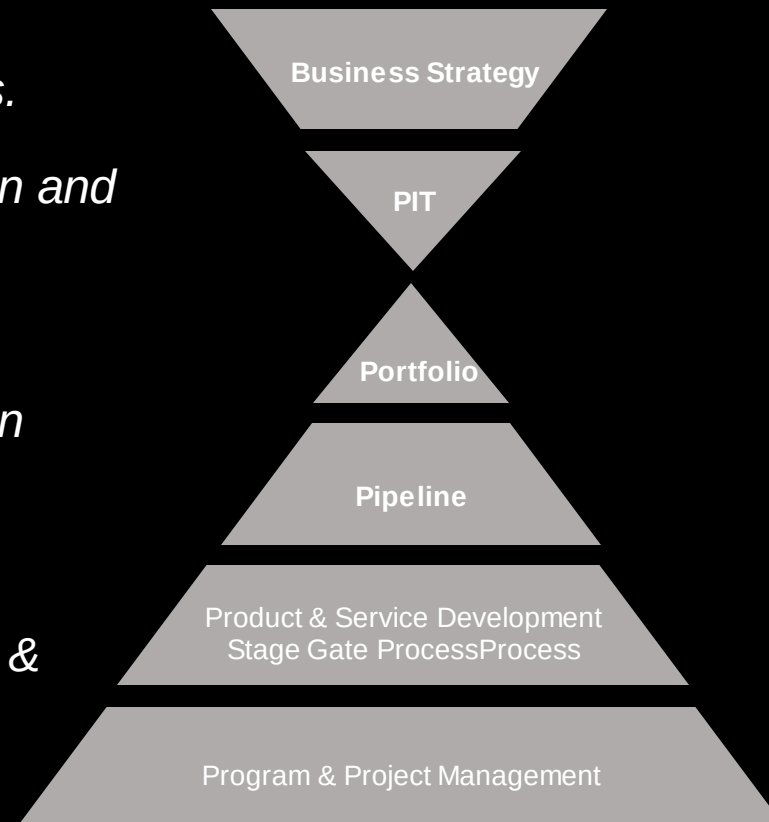


PRODUCT AND SERVICE LIFECYCLE



PRODUCT & SERVICE DEVELOPMENT

- *From strategy, all else flows.*
- *Product & service innovation and development is part of the business strategy.*
- *Product & service innovation drives the Portfolio & development pipeline*
- *Multiple ways to do product & service development*



Business Strategy

Defines the direction of the company

Product Innovation & Tech Strategy (PIT)

Defines the direction of the company

Portfolio

Operationalises the product innovation & technology strategy

Pipeline

Prioritises programs & projects to allocate resources (time, money, people)

Stage gate process

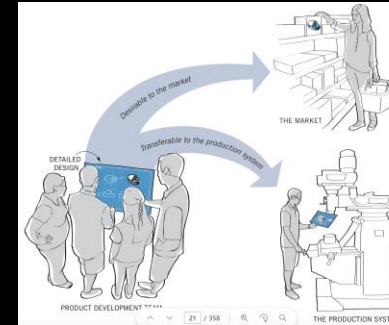
Guides and directs the innovation efforts from idea through launch

Program & Project Management

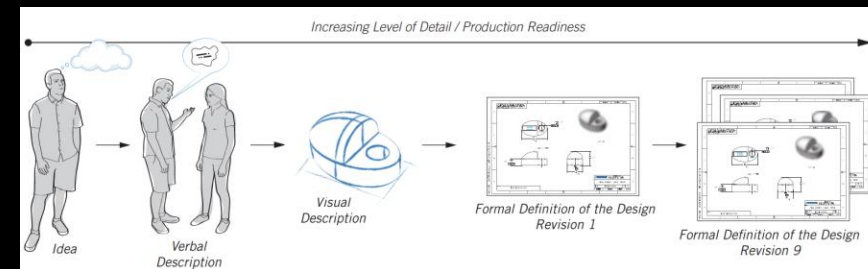
Manages the budgets and milestones of individual project.

FOUR PRINCIPLES OF PRODUCT DEVELOPMENT

Principle 1: The goal of the product development team is a design that desirable and transferrable (design vs. production system).

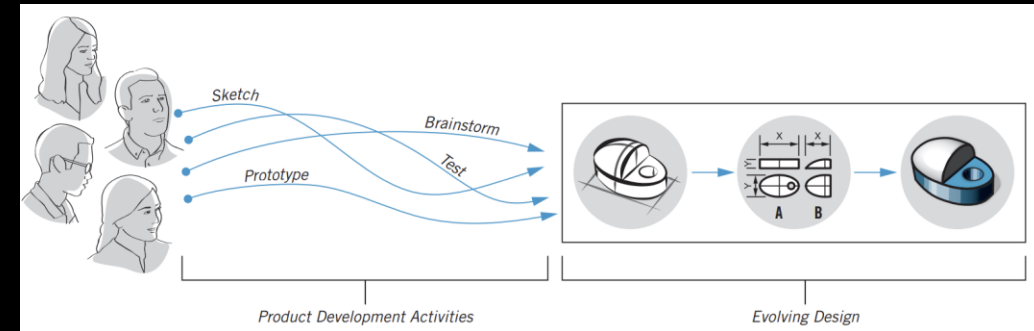


Principle 2: The design must evolve gradually getting better, mature to contain all the information for production, sales, marketing & services.

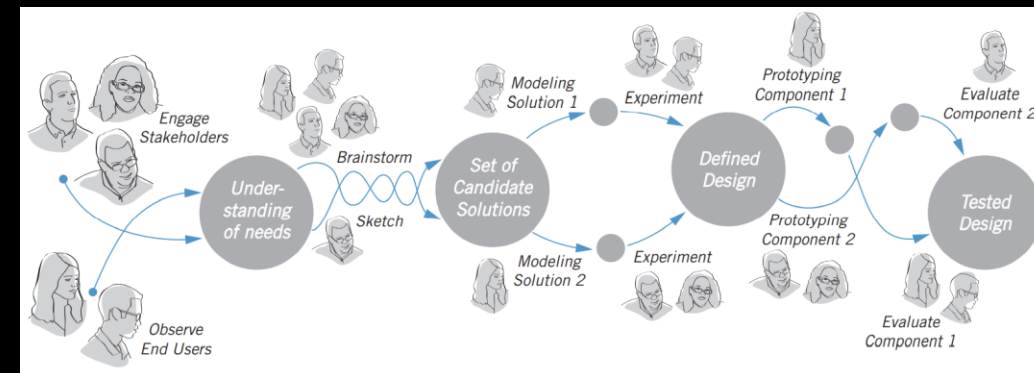


FOUR PRINCIPLES OF PRODUCT DEVELOPMENT

Principle 3: The product development team drives the evolution of the design through activities that create artifacts.



Principle 4: The optimum evolution of the product requires customisation and coordination of activities.

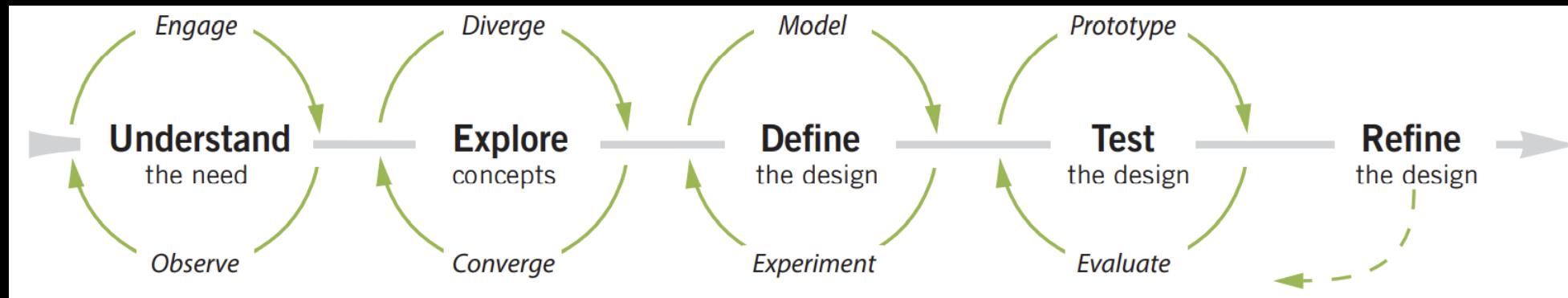




BASIC DESIGN PROCESS

1. Understand the need: Understand the problem & opportunity to be solved. Identify the who has (persona) the problem. Ask and observe to collect the requirements.
2. Explore Concepts: From the requirements formulate concepts that can solve the problem. This out-side-of-the-box (TRIZ)
3. Define the Design: Select the winning concept and add engineering detail to evaluate its performance and behaviour.
4. Test the Design: Use digital twins, hybrid twins and real-world twins to predict how the design will operate in the real world.
5. Refine the Design: the design will be improved through identified problems and ideas that will result in changes requests CR).

BASIC DESIGN PROCESS



Engage with the user and stakeholder to understand.

Observe how the user & customer use products to collect requirements.

Diverge to select concepts that are unique and competitive.

Converge the number of concepts to a manageable number (where to invest?)

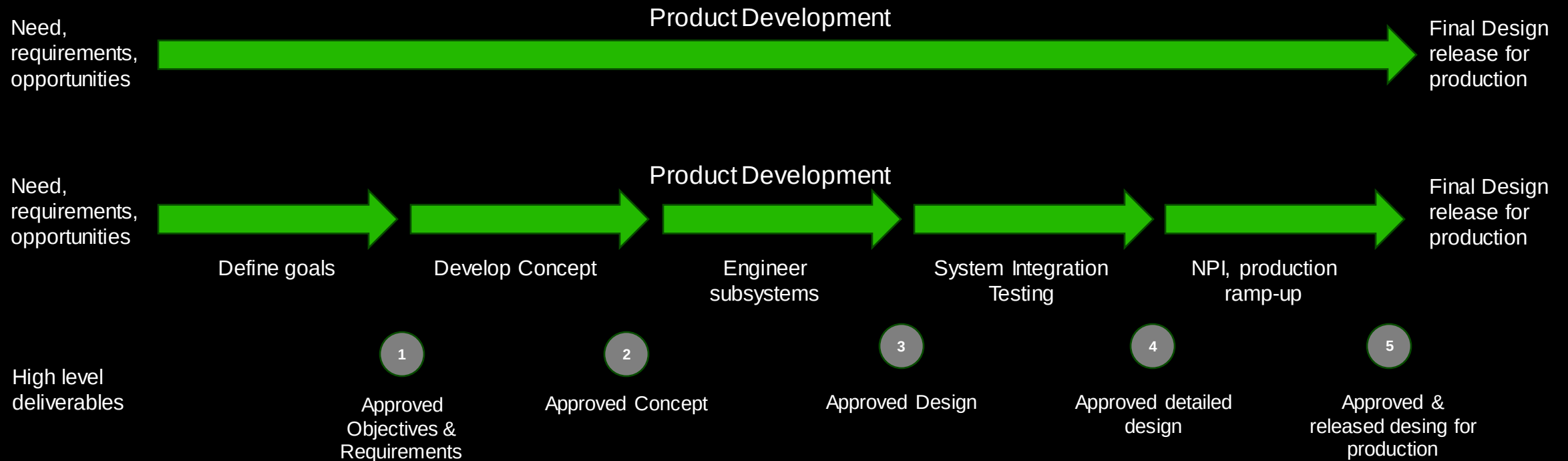
Model the product to add detail (material, dimensions standard parts).

Experiment with the design virtually and explore improvement possibilities

Prototype with real-world examples or real-time multibody physics simulations

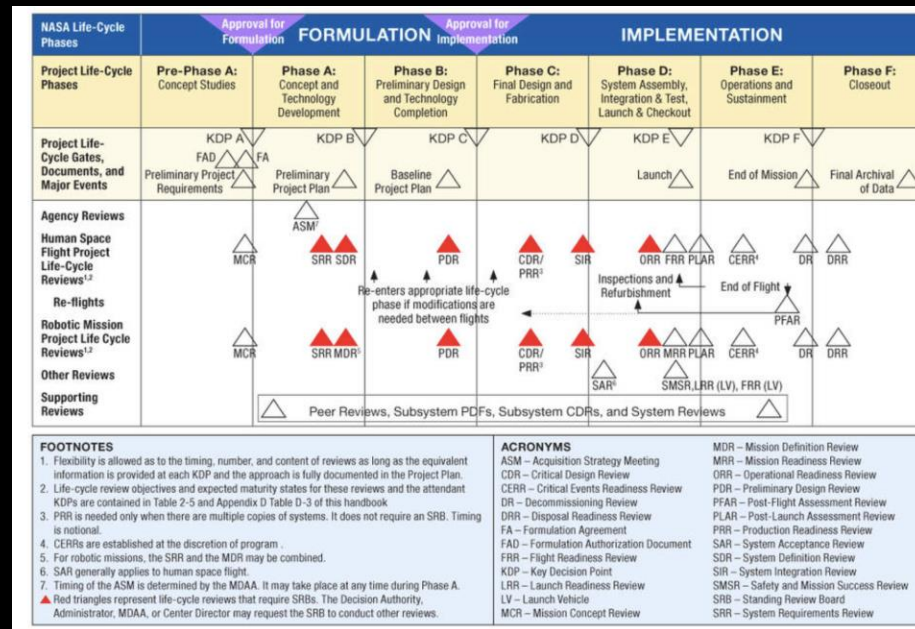
Evaluate if the design fulfils the market and production needs (quantitative & qualitative methods)

PRODUCT DEVELOPMENT ACTIVITY MAP



DEVELOPMENT PROCESS EXAMPLES

Waterfall Development Model Example



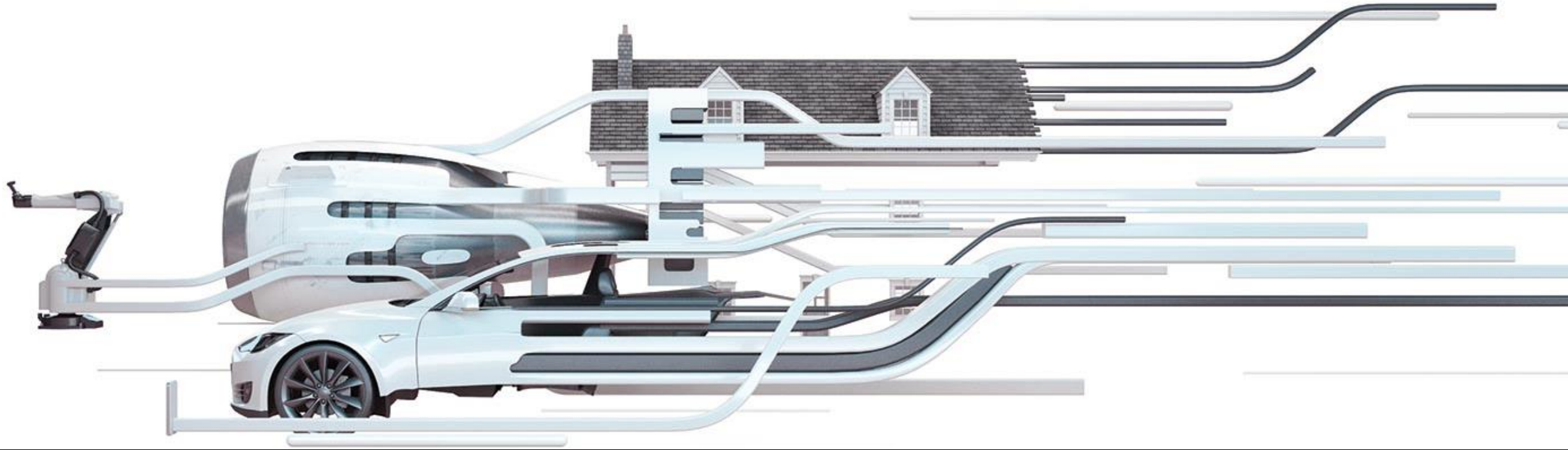
Front End Loading (FEL) Front-End Engineering Design (FEED)

PHASE	FEL I	FEL II	FEL III	A P P R O V A L	FEL IV
Project Phase	Planning / Conceptual Design	Feasibility Design	Preliminary Engineering		Detail Engineering
Engineering Definition	2% to 5%	10% to 30%	30% to 60%		60% to 90%
AACE Estimate Class	Class 5	Class 4	Class 3		Class 2
AACE Cost Estimate Accuracy	L: -20% to -50% H: +30% to +100%	L: -15% to -30% H: +20% to +50%	L: -10% to -20% H: +10% to +30%		L: -5% to -15% H: +5% to +20%
Basis of Estimate	Parametric models, factors, experience / judgement	Budgetary quotes, factors, AE estimates	Semi-detailed unit costs, key contract RFPs		Detailed take-offs / unit costs, quoted contracts





ANY
QUESTIONS
?



WHY IS PRODUCT & SERVICE DEVELOPMENT IMPORTANT?

INNOVATION ECONOMIES DRIVE TRANSFORMATION



SOME KEY FINDINGS

The GII 2021 finds that investment in innovation has shown **great resilience** during the COVID-19 pandemic, often **reaching new peaks**, but that it varies across sectors and regions

New S&T (science and technology) **clusters** are emerging, with the majority located in only a handful of countries

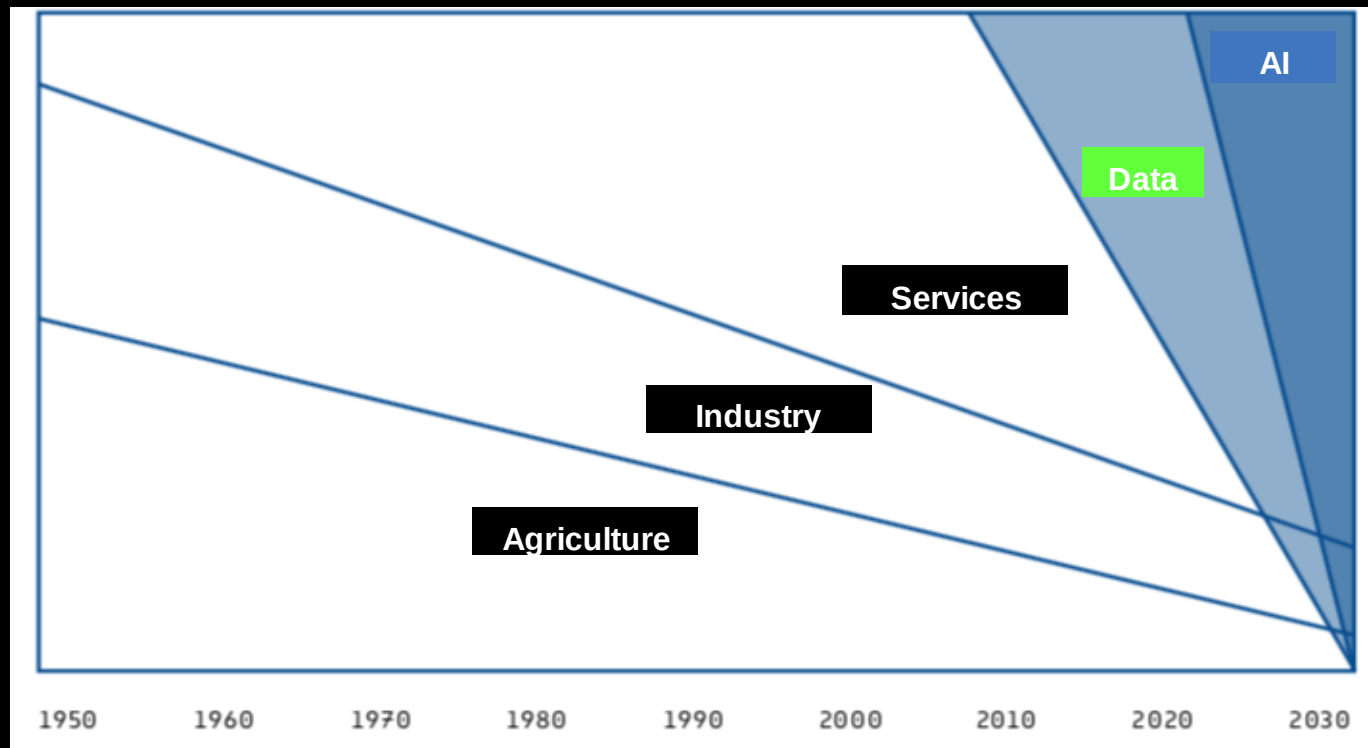
Technological progress at the frontier holds substantial promise

Only a few economies have **consistently delivered** peak innovation performance

INNOVATION ECONOMIES DRIVE TRANSFORMATION

Country/Economy	Overall GII	Institutions	Human capital and research	Infrastructure	Market sophistication	Business sophistication	Knowledge and technology outputs	Creative outputs
Switzerland	1	13	6	2	6	4	1	2
Sweden	2	9	2	3	11	1	2	5
United States of America	3	12	11	23	2	2	3	12
United Kingdom	4	15	10	10	4	21	10	4
Republic of Korea	5	28	1	12	18	7	8	8
Netherlands	6	6	14	16	31	5	7	7
Finland	7	2	4	11	19	6	5	16
Singapore	8	1	9	15	5	3	13	17
Denmark	9	8	5	5	7	11	14	13
Germany	10	17	3	21	20	12	9	11
France	11	19	15	17	17	19	16	6
China	12	61	21	24	16	13	4	14
Japan	13	7	20	9	15	10	11	18
Hong Kong, China	14	11	25	6	3	24	62	1
Israel	15	34	19	40	8	8	6	30

PREDICTION OF FINNISH NATIONAL ECONOMIES TRANSFORMATION



Questions to report

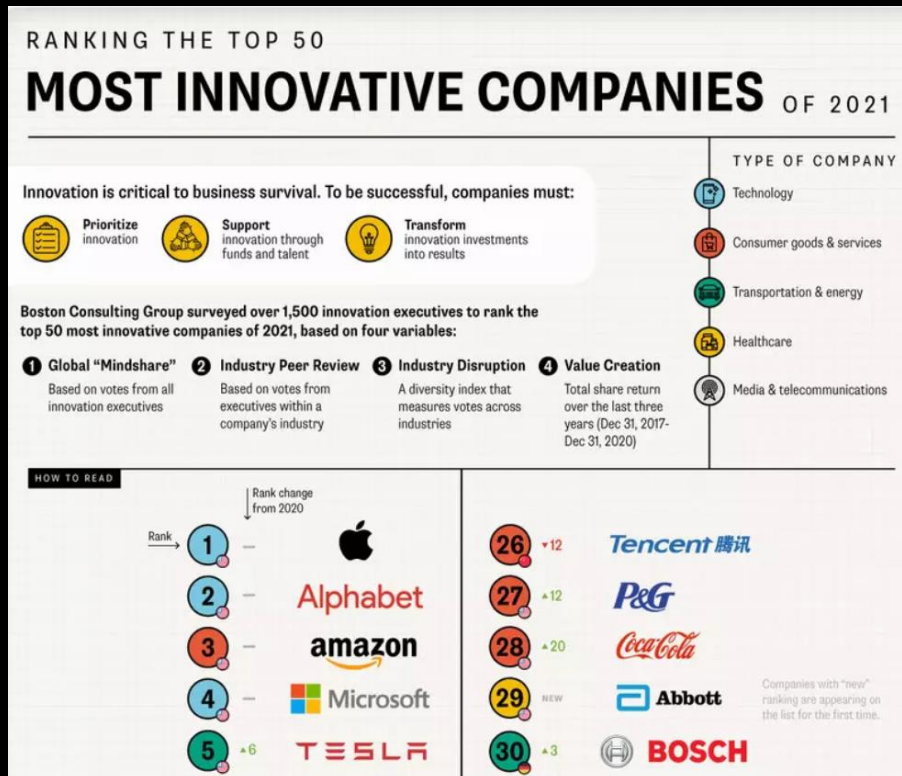
➤ iPSS (SCPSS) potential is unclear from report

10 TYPES OF INNOVATION

- Innovation drives business forward, but it is often misunderstood.
- After examining over 2,000 innovations throughout history, Doblin discovered most breakthroughs don't stem from engineering inventions or rare discoveries.
- In reality, there are 10 different types of innovation

#	Innovation Type	Description
1.	Profit Model	How you make money
2.	Network	Connections with others to create value
3.	Structure	Alignment of your talent and assets
4.	Process	Signature of superior methods for doing your work
5.	Product Performance	Distinguishing features and functionality
6.	Product System	Complementary products and services
7.	Service	Support and enhancements that surround your offerings
8.	Channel	How your offerings are delivered to customers and users
9.	Brand	Representation of your offerings and business
10.	Customer Engagement	Distinctive interactions you foster

TOP 50 MOST INNOVATIVE COMPANIES

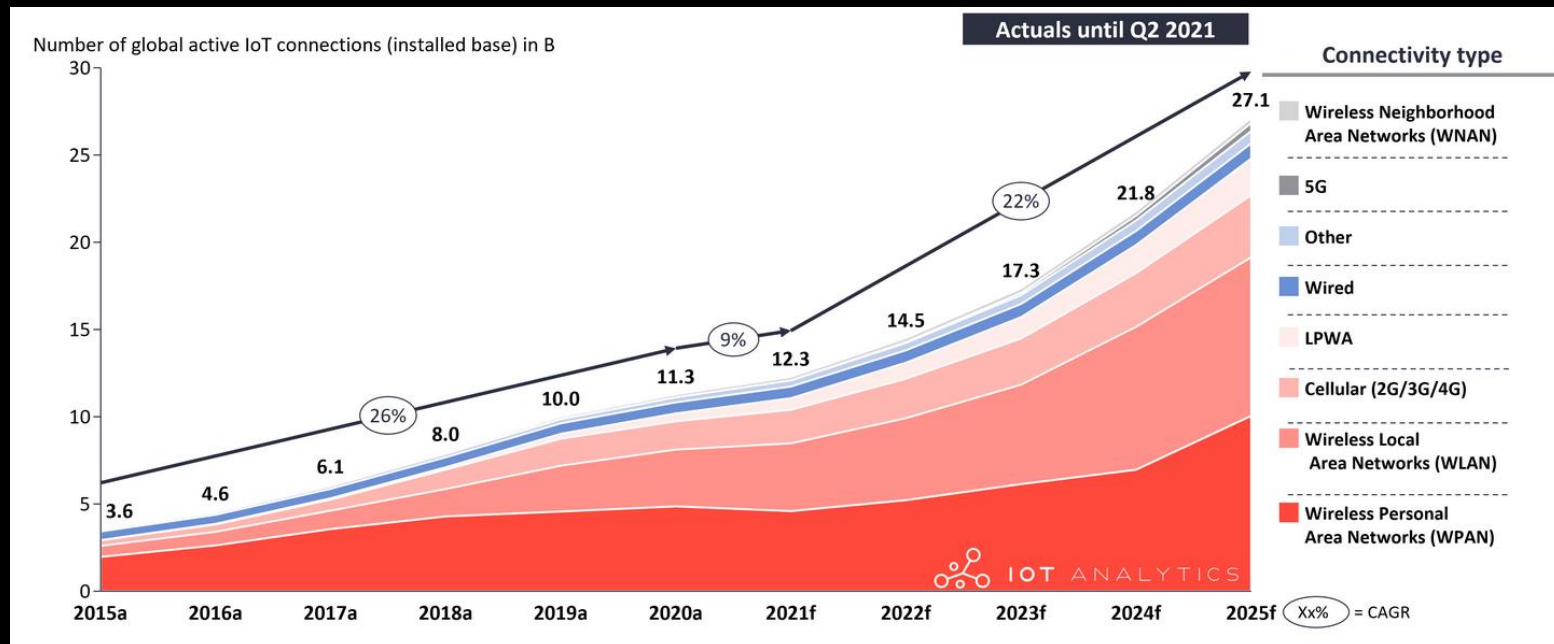


SOME KEY FINDINGS

- » Pure B2B manufacturing companies missing from list
 - » Tesla (5)
 - » Siemens (11)
 - » Bosch (3)
- » Platform & software+ companies dominate
- » 10 Healthcare companies (5 previous year)

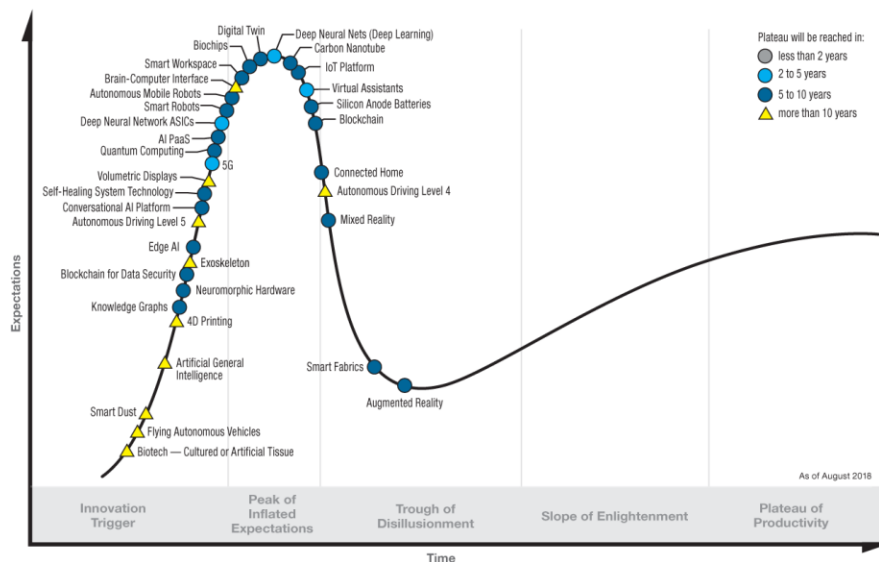
GLOBAL IOT MARKET FORECAST (BILLION DEVICES)

- » Today a large amount of unconnected devices are industrial and offer growth potential
- » Challenge is brown field PSS vs. green field iPSS



BUSINESS VS TECHNICAL CAPABILITY

Hype Cycle for Emerging Technologies, 2018



gartner.com/SmarterWithGartner

Source: Gartner (August 2018)
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Gartner.

SOME THINGS TO KEEP IN MIND

- » Change business, society and the environment
- » Balance the customer & segment needs
- » iPSS Innovation vs. efficiency
- » Business models, Value Proposition
- » Apply novel technologies with best ROI



ANY
QUESTIONS
?

